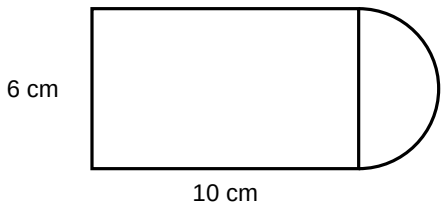
Review Sheet — Stoichiometry and Geometry

Name: _____ Date: _____ Score: _____ / 30

Show your reasoning. Molar masses: H = 1.0 g/mol, O = 16.0 g/mol. Use the exact value of π.

1. Hydrogen burns in oxygen according to $2H_2 + O_2 \rightarrow 2H_2O$. How many grams of water are produced from **8.0 g** of H_2 with excess oxygen? (10 pts)

2. The figure shows a rectangle 10 cm long and 6 cm wide with a semicircle attached to its right side. Find the total area of the figure. (10 pts)



3. A right triangle has legs of length 6 cm and 8 cm. Find the length of its hypotenuse. (10 pts)